Example 3.37 Simplify $\dfrac{\sin 75^\circ - \sin 15^\circ}{\cos 75^\circ + \cos 15^\circ}$

Solution:

$$\text{We have } \frac{\sin 75^\circ - \sin 15^\circ}{\cos 75^\circ + \cos 15^\circ} = \frac{2\cos\left(\frac{75^\circ + 15^\circ}{2}\right)\sin\left(\frac{75^\circ - 15^\circ}{2}\right)}{2\cos\left(\frac{75^\circ + 15^\circ}{2}\right)\cos\left(\frac{75^\circ - 15^\circ}{2}\right)}$$

$$= \frac{2\cos 45^\circ \sin 30^\circ}{2\cos 45^\circ \cos 30^\circ} = \tan 30^\circ = \frac{1}{\sqrt{3}}.$$

Try to solve using $\sin 75^\circ = \cos 15^\circ$ and $\cos 75^\circ = \sin 15^\circ$

Example 3.38 Show that $\cos 10^\circ \cos 30^\circ \cos 50^\circ \cos 70^\circ = \dfrac{3}{16}$.

Solution:

We know that $\cos(60^\circ - A)\cos A \cos(60^\circ + A) = \dfrac{1}{4}\cos 3A$

$$\begin{aligned}\cos 10^\circ \cos 30^\circ \cos 50^\circ \cos 70^\circ &= \cos 30^\circ \left[\cos 10^\circ \cos 50^\circ \cos 70^\circ\right] \\ &= \cos 30^\circ \left[\cos(60^\circ - 10^\circ)\cos 10^\circ \cos(60^\circ + 10^\circ)\right] \\ &= \frac{\sqrt{3}}{2}\left[\frac{1}{4}\cos 30^\circ\right] = \frac{\sqrt{3}}{2}\left(\frac{1}{4}\right)\left(\frac{\sqrt{3}}{2}\right) = \frac{3}{16}\end{aligned}$$

Exercise - 3.6

1. Express each of the following as a sum or difference

   (i) $\sin 35^\circ \cos 28^\circ$ (ii) $\sin 4x \cos 2x$ (iii) $2\sin 10\theta \cos 2\theta$ (iv) $\cos 5\theta \cos 2\theta$ (v) $\sin 5\theta \sin 4\theta$.

2. Express each of the following as a product

   (i) $\sin 75^\circ - \sin 35^\circ$ (ii) $\cos 65^\circ + \cos 15^\circ$ (iii) $\sin 50^\circ + \sin 40^\circ$ (iv) $\cos 35^\circ - \cos 75^\circ$.

3. Show that $\sin 12^\circ \sin 48^\circ \sin 54^\circ = \dfrac{1}{8}$.

4. Show that $\cos\dfrac{\pi}{15}\cos\dfrac{2\pi}{15}\cos\dfrac{3\pi}{15}\cos\dfrac{4\pi}{15}\cos\dfrac{5\pi}{15}\cos\dfrac{6\pi}{15}\cos\dfrac{7\pi}{15} = \dfrac{1}{128}$.

5. Show that $\dfrac{\sin 8x \cos x - \sin 6x \cos 3x}{\cos 2x \cos x - \sin 3x \sin 4x} = \tan 2x$.

6. Show that $\dfrac{(\cos\theta - \cos 3\theta)(\sin 8\theta + \sin 2\theta)}{(\sin 5\theta - \sin\theta)(\cos 4\theta - \cos 6\theta)} = 1$.

7. Prove that $\sin x + \sin 2x + \sin 3x = \sin 2x\,(1 + 2\cos x)$.

8. Prove that $\dfrac{\sin 4x + \sin 2x}{\cos 4x + \cos 2x} = \tan 3x$.

9. Prove that $1 + \cos 2x + \cos 4x + \cos 6x = 4\cos x \cos 2x \cos 3x$.